

Dataset Research for SIH26038

Explainable AI-Based Diabetic Retinopathy Screening

Problem Statement: SIH26038
Domain: Healthcare / Medical Computer Vision / Explainable AI
Objective: Develop an explainable AI-based retinal screening pipeline capable of assessing image quality, detecting retinal abnormalities, grading diabetic retinopathy, providing visual explanations, and supporting scalable telemedicine screening.

1. Research Objective

SIH26038 requires more than a simple diabetic retinopathy classification model. The proposed system needs to address multiple tasks:

- Retinal image quality assessment
- Image enhancement and preprocessing
- Diabetic retinopathy severity grading
- Retinal lesion detection/segmentation
- Optic disc and fovea localization
- Blood-vessel segmentation
- Explainable AI using techniques such as Grad-CAM
- Confidence estimation
- External validation
- Scalability analysis for rural telemedicine

No single publicly available dataset provides all of these annotations together.

Therefore, a **multi-dataset strategy** is required.

2. Dataset Selection Strategy

Dataset	Primary Purpose	Role in SIH26038
APTOS 2019	DR severity classification	Main 5-class DR classifier
IDRiD	Indian DR + lesion annotations	Indian-domain adaptation + lesion/XAI validation
DRIVE	Retinal vessel segmentation	Blood-vessel segmentation
Messidor-2	Independent retinal dataset	External validation
DDR	Large-scale DR + lesion data	Optional large-scale training

Dataset	Primary Purpose	Role in SIH26038
EyePACS	Large-scale DR classification	Optional pretraining
e-optha	Lesion-specific datasets	Optional lesion detector training

The recommended core dataset combination is:

APTOS + IDRiD + DRIVE + Messidor-2

The remaining datasets can be incorporated if additional training data or specialized lesion annotations are required.

3. APTOS 2019

3.1 Overview

APTOS 2019 Blindness Detection is a diabetic retinopathy classification dataset released as part of the APTOS 2019 Kaggle competition.

It contains **3,662 labelled retinal fundus images**.

Each image is assigned one of five DR grades:

Label	Severity
0	No DR
1	Mild DR
2	Moderate DR
3	Severe DR
4	Proliferative DR

3.2 Main Use

APTOS will be used as the **primary dataset for training the DR severity classification model**.

Pipeline

```

APTOS Images
  ↓
Preprocessing

```

↓
Data Augmentation
↓
Transfer Learning
↓
5-Class Classifier
↓
DR Grade 0–4

3.3 Why APTOS?

APTOS is highly suitable because the SIH problem requires five-level DR grading.

It also contains real-world variations in retinal image quality, making it useful for developing robust preprocessing and quality-control mechanisms.

3.4 Advantages

- 5-class DR labels
- Suitable for transfer learning
- Real-world retinal photographs
- Commonly used benchmark
- Useful for classification experiments

3.5 Limitations

- Significant class imbalance
- Does not provide comprehensive pixel-level lesion annotations
- Not sufficient for lesion segmentation
- Dataset domain may differ from rural screening environments

3.6 Planned Use

Primary: DR severity classifier

Evaluation: Accuracy, Macro-F1, Sensitivity, Specificity, Balanced Accuracy, AUROC and Quadratic Weighted Kappa.

Source

Kaggle — APTOS 2019 Blindness Detection
<https://www.kaggle.com/competitions/aptos2019-blindness-detection/data>

4. IDRiD

4.1 Overview

IDRiD — Indian Diabetic Retinopathy Image Dataset is one of the most important datasets for this project because it contains retinal images from an Indian clinical setting.

The dataset contains **516 retinal fundus images** with image-level DR/DME grading.

The images were acquired at an eye clinic in **Nanded, Maharashtra, India**.

4.2 Available Information

IDRiD provides:

- DR severity grading
- Diabetic macular edema grading
- Optic disc information
- Fovea information
- Lesion annotations
- Pixel-level segmentation masks for selected images

4.3 DR Grading

Grade	Description
0	No DR
1	Mild NPDR
2	Moderate NPDR
3	Severe NPDR
4	Proliferative DR

4.4 Lesion Annotations

IDRiD provides pixel-level annotations for clinically relevant lesions.

Lesion	Approx. Annotated Images
Microaneurysms	81
Hard Exudates	81
Hemorrhages	80

Lesion	Approx. Annotated Images
Soft Exudates	40
Optic Disc	81

4.5 Main Use

IDRiD will have **three major roles**:

1. Indian-domain adaptation

```

APTOS-trained model
    ↓
  IDRiD
    ↓
Fine-tuning / adaptation
    ↓
Indian-domain model
  
```

2. Lesion segmentation

```

IDRiD
    ↓
Annotated lesion masks
    ↓
Segmentation model
    ↓
Microaneurysm / Exudate /
Hemorrhage / Soft Exudate
  
```

3. Explainable AI validation

```

DR Classifier
    ↓
Grad-CAM
    ↓
Attention Map
    ↓
Compare with IDRiD Lesion Mask
    ↓
Quantitative XAI Evaluation
  
```

4.6 Why IDRiD is particularly important

The SIH problem is focused on healthcare deployment in India.

IDRiD provides Indian retinal data and detailed lesion annotations, making it highly valuable for evaluating whether the model focuses on clinically relevant retinal regions.

4.7 Limitations

- Only 516 images
- Pixel-level lesion annotations are available for a much smaller subset
- Limited geographic diversity
- Not sufficient as the only training dataset for a deep-learning model

4.8 Planned Use

Primary: Indian-domain adaptation

Secondary: Lesion segmentation

Secondary: XAI validation

Secondary: Optic disc/fovea localization

Source

IDRiD Official Dataset

<https://idrid.grand-challenge.org/Data/>

5. DRIVE

5.1 Overview

DRIVE — Digital Retinal Images for Vessel Extraction is a retinal blood-vessel segmentation dataset.

It contains **40 retinal images**:

- 20 training images
- 20 test images

The dataset includes manually annotated retinal vessel segmentation masks.

5.2 Main Use

DRIVE will be used specifically for **blood-vessel segmentation**.

```
DRIVE Image
  ↓
Preprocessing
  ↓
U-Net / Segmentation Model
  ↓
Retinal Vessel Mask
```

5.3 Why DRIVE?

The SIH problem requires analysis of retinal structures, including blood vessels.

DRIVE provides manually annotated vessel masks and is therefore appropriate for developing and benchmarking a vessel segmentation component.

5.4 Possible Model

A U-Net-based architecture can be used:

```
Fundus Image
  ↓
Encoder
  ↓
Feature Extraction
  ↓
Decoder
  ↓
Vessel Segmentation Mask
```

5.5 Limitations

- Only 40 images
- Not suitable for training a complete DR classifier
- Primarily designed for vessel segmentation
- Different population/domain from Indian datasets

5.6 Planned Use

Primary: Retinal blood-vessel segmentation

Not recommended for: 5-class DR classification

Source

DRIVE Official Dataset

<https://drive.grand-challenge.org/DRIVE/>

6. Messidor-2

6.1 Overview

Messidor-2 is an independent retinal-image dataset containing approximately **1,748 images from 874 examinations**.

It is useful as an external dataset because it comes from a different data collection environment than APTOS and IDRiD.

6.2 Main Use

Messidor-2 will primarily be used for **external validation**.

```
APTOS
  ↓
Train Model
  ↓
Freeze Model
  ↓
Messidor-2
  ↓
External Evaluation
```

6.3 Why External Validation?

A model can perform very well on its training domain but perform poorly on images from another camera, hospital or population.

Therefore:

```
High APTOS performance
  ≠
Guaranteed real-world robustness
```

External testing helps measure generalization.

6.4 Important Limitation

The official Messidor-2 distribution does not provide an official DR ground-truth annotation set in the same way as datasets such as APTOS or IDRiD.

Therefore, any third-party annotations must be clearly identified and not presented as official Messidor-2 ground truth.

6.5 Planned Use

Primary: External validation / generalization testing

Source

ADCIS — Messidor-2

<https://www.adcis.net/en/third-party/messidor2/>

7. DDR Dataset

7.1 Overview

DDR — Diabetic Retinopathy Dataset is a larger retinal dataset designed for diabetic retinopathy research.

A commonly used version contains approximately **13,673 retinal images**.

It supports research involving:

- DR classification
- Lesion detection
- Lesion segmentation

7.2 Main Use

DDR can be used when additional training data are required.

```
APTOS
+
DDR
+
Optional EyePACS
```

↓
Large-scale training
↓
IDRiD adaptation

7.3 Advantages

- Much larger than IDRiD
- Useful for lesion-related research
- Supports multiple DR tasks
- Useful for improving generalization

7.4 Limitations

- Not specifically focused on Indian demographics
- Additional preprocessing and dataset harmonization required
- Larger computational requirement

7.5 Planned Use

Optional: Large-scale DR and lesion training

Source

DDR Dataset Repository
<https://github.com/nkicli/DDR-dataset>

8. EyePACS

8.1 Overview

EyePACS is one of the largest commonly used retinal-image datasets for DR classification.

A widely used labelled training collection contains approximately **35,126 images**.

8.2 Main Use

EyePACS can be used for:

- Large-scale pretraining
- Transfer learning
- DR classification
- Robustness experiments

```
EyePACS
↓
Large-scale pretraining
↓
APTOS
↓
IDRiD adaptation
```

8.3 Advantages

- Large dataset
- Five-grade DR classification
- Useful for deep-learning pretraining

8.4 Limitations

- Large computational/storage requirements
- Different domain from Indian clinical data
- Does not provide the detailed lesion masks available in IDRiD

8.5 Planned Use

Optional: Large-scale pretraining

9. e-optha

9.1 Overview

e-optha provides specialized retinal lesion datasets.

Important subsets include datasets for:

- Exudate detection
- Microaneurysm/small hemorrhage detection

9.2 Main Use

e-optha can be used to improve specialized lesion detectors.

```
e-optha
↓
Lesion-specific training
```

↓
Microaneurysm detector
+
Exudate detector

9.3 Advantages

- Specialized lesion annotations
- Useful for lesion detection research

9.4 Limitations

- Smaller than large-scale DR classification datasets
- Not appropriate as the primary 0–4 DR classification dataset

Source

ADCIS — e-optha

<https://www.adcis.net/en/third-party/e-optha/>

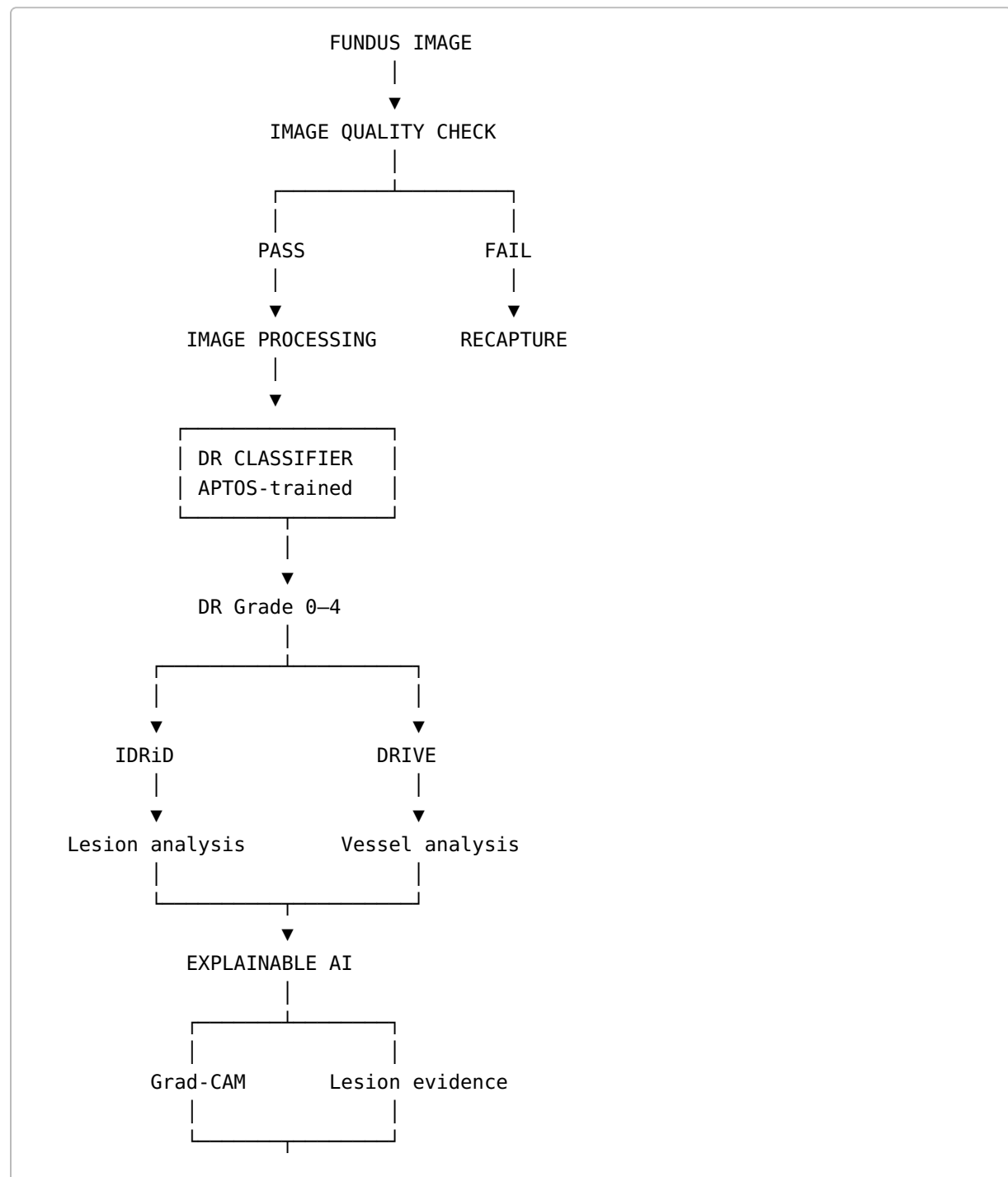
10. Dataset-to-Requirement Mapping

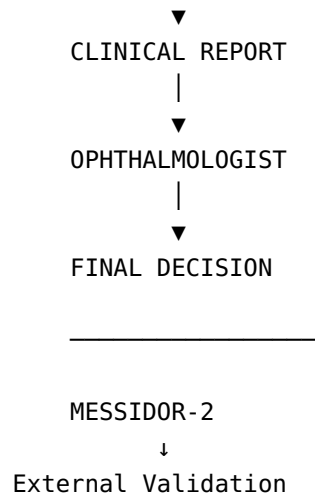
SIH26038 Requirement	Dataset	Proposed Method
DR Grade 0–4	APTOS	Classification
Indian demographic	IDRiD	Domain adaptation
Microaneurysms	IDRiD / e-optha	Detection/segmentation
Hard exudates	IDRiD / e-optha	Detection/segmentation
Hemorrhages	IDRiD	Segmentation
Soft exudates	IDRiD	Segmentation
Optic disc	IDRiD	Localization/segmentation
Fovea	IDRiD	Localization
Blood vessels	DRIVE	Segmentation
Explainability	IDRiD	Grad-CAM vs lesion masks
External validation	Messidor-2	Generalization testing
Large-scale training	EyePACS / DDR	Optional
Image quality	APTOS + custom quality module	Quality assessment

11. Recommended Dataset Architecture

The datasets should **not** simply be merged into one giant dataset.

Instead, each dataset should serve a specific purpose.





12. Recommended Training Strategy

Stage 1 — Base DR Classification

Dataset:

APTOS 2019

Task:

5-class DR classification

Output:

0 → No DR
1 → Mild
2 → Moderate
3 → Severe
4 → Proliferative

Stage 2 — Indian Domain Adaptation

Dataset:

IDRiD

Approach:

```
graph TD; A[APTOS-trained model] --> B[IDRiD fine-tuning]; B --> C[Indian-domain adapted model];
```

The purpose is not to claim that IDRiD represents all Indian patients, but to evaluate and adapt the model toward an Indian retinal-image domain.

Stage 3 — Lesion Segmentation

Dataset:

IDRiD

Possible architecture:

U-Net / U-Net variants

Target lesions:

- Microaneurysms
 - Hard exudates
 - Hemorrhages
 - Soft exudates
-

Stage 4 — Vessel Segmentation

Dataset:

DRIVE

Possible architecture:

U-Net

Output:

```
Fundus image
  ↓
Vessel segmentation
  ↓
Binary vessel mask
```

Stage 5 — Explainable AI

Use:

Grad-CAM

Pipeline:

```
Fundus Image
  ↓
DR Classifier
  ↓
Predicted Grade
  ↓
Grad-CAM
  ↓
Attention Map
  ↓
Compare with IDRiD lesion annotations
```

Possible evaluation:

- IoU
- Dice coefficient
- Pointing accuracy
- Localization metrics

This provides quantitative evidence that the model is focusing on clinically relevant regions rather than only producing visually attractive heatmaps.

Stage 6 — External Validation

Dataset:

Messidor-2

The final model should be frozen before external evaluation.

```
Training
APTOS / optional DDR / EyePACS
↓
Validation
IDRiD
↓
MODEL FREEZED
↓
External Test
Messidor-2
```

This helps evaluate cross-dataset generalization.

13. Dataset Leakage Prevention

Dataset leakage must be avoided.

The following approach is recommended:

```
TRAINING
APTOS
+
Optional DDR/EyePACS
↓
MODEL DEVELOPMENT
↓
IDRiD validation/adaptation
↓
MODEL FREEZE
↓
MESSIDOR-2
↓
FINAL EXTERNAL EVALUATION
```

The external test dataset should not be repeatedly used to tune the model.

14. Class Imbalance

APTOS and other DR datasets have imbalanced classes.

Therefore, ordinary accuracy should not be the only evaluation metric.

Recommended metrics:

Classification

- Accuracy
- Macro-F1
- Sensitivity
- Specificity
- Balanced Accuracy
- AUROC
- Confusion Matrix
- Quadratic Weighted Kappa

Segmentation

- Dice Score
- IoU
- Sensitivity
- Specificity

Explainability

- Lesion/attention overlap
- IoU
- Dice
- Localization accuracy

15. Image Quality Assessment

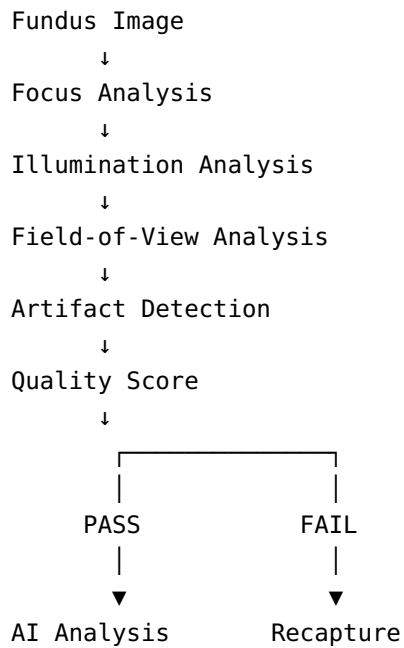
SIH26038 requires the system to evaluate:

- Focus
- Illumination
- Field of view

There is no need to assume that one existing DR dataset provides perfect ground-truth quality labels for all three characteristics.

Instead, a dedicated quality-control module can be developed.

Proposed pipeline



Possible image-quality features include:

- Blur/focus metrics
- Brightness statistics
- Contrast
- Retinal field coverage
- Black-border ratio
- Artifact detection

A quality threshold should be validated experimentally rather than arbitrarily chosen.

16. Why We Need Multiple Datasets

A single dataset cannot satisfy every requirement.

APTOS

Answers:

"What is the DR severity?"

IDRiD

Answers:

"What lesions are present, and how does the model behave on Indian retinal images?"

DRIVE

Answers:

"Where are the retinal blood vessels?"

Messidor-2

Answers:

"Does the model generalize to an independent dataset?"

DDR / EyePACS

Answers:

"Can we increase the amount of training data?"

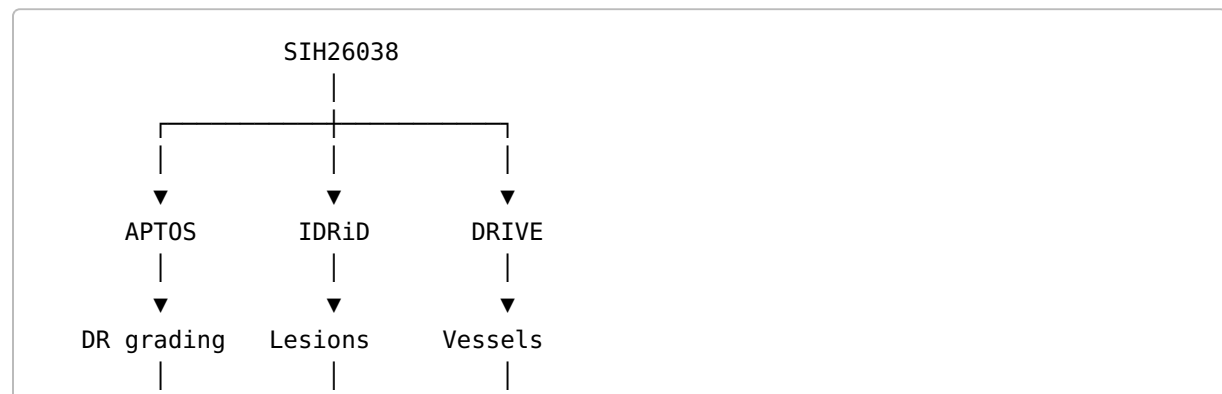
e-optha

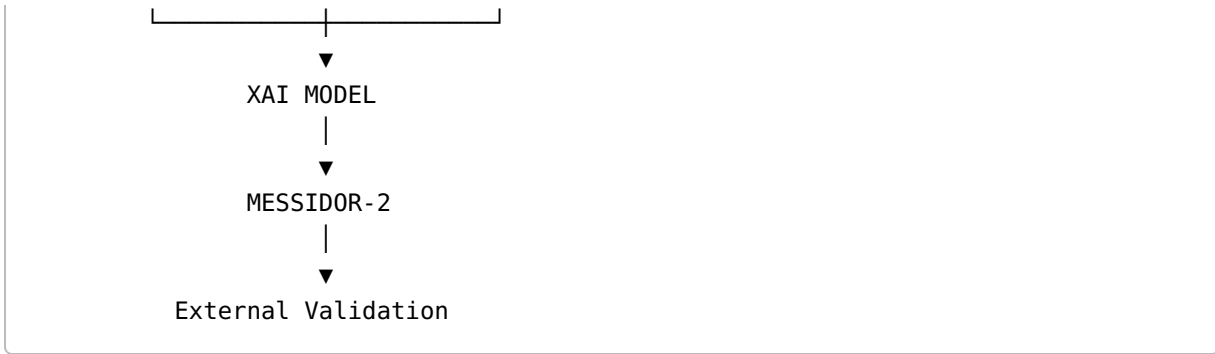
Answers:

"Can specialized lesion detectors be improved?"

17. Core Dataset Recommendation

For the initial SIH prototype, the recommended minimum dataset stack is:





Priority

1. APTOS — MUST HAVE
2. IDRiD — MUST HAVE
3. DRIVE — MUST HAVE
4. Messidor-2 — STRONGLY RECOMMENDED
5. DDR — OPTIONAL
6. EyePACS — OPTIONAL
7. e-optha — OPTIONAL

18. Dataset Limitations and Research Gaps

Despite the availability of multiple datasets, several challenges remain.

1. Domain shift

Different datasets come from different:

- Countries
- Hospitals
- Cameras
- Imaging protocols
- Patient populations

Therefore, a model trained on one dataset may not perform identically on another.

2. Limited Indian data

IDRiD is highly valuable but contains only 516 images.

It should not be treated as a complete representation of the Indian population.

3. Limited lesion annotations

Detailed pixel-level annotations are available for only a subset of IDRiD.

4. Image quality labels

There is no single perfect public dataset matching all SIH image-quality requirements.

5. Class imbalance

Severe and proliferative DR classes are typically much smaller than the no-DR class.

6. Clinical deployment

Public datasets are useful for research but cannot replace prospective clinical validation.

19. Final Dataset Strategy

The proposed research strategy is:

Train the primary DR severity classifier using APTOS 2019, use IDRiD for Indian-domain adaptation and lesion-level explainability, use DRIVE for retinal blood-vessel segmentation, and use Messidor-2 as an independent external validation dataset. DDR and EyePACS may be incorporated for additional large-scale training, while e-optha can be used for specialized lesion detection.

This creates a modular architecture rather than forcing unrelated datasets into one training set.

20. Final Recommendation

Core Research Stack

Dataset	Priority	Task
APTOS 2019	● Essential	DR Grade 0–4
IDRiD	● Essential	Indian data + lesions + XAI

Dataset	Priority	Task
DRIVE	● Essential	Vessel segmentation
Messidor-2	● Recommended	External validation
DDR	● Optional	Large-scale DR/lesion training
EyePACS	● Optional	Large-scale pretraining
e-ophtha	● Optional	Specialized lesion detection

Recommended first implementation

```

APTOS
  ↓
DR Severity Classifier
  ↓
IDRiD
  ├── Indian adaptation
  ├── Lesion segmentation
  └── XAI validation
  ↓
DRIVE
  └── Vessel segmentation
  ↓
Final Screening Pipeline
  ↓
Messidor-2
  └── External validation

```

21. References

1. **APTOS 2019 Blindness Detection — Kaggle**
<https://www.kaggle.com/competitions/aptos2019-blindness-detection/data>
2. **Indian Diabetic Retinopathy Image Dataset (IDRiD)**
<https://idrid.grand-challenge.org/Data/>
3. **DRIVE — Digital Retinal Images for Vessel Extraction**
<https://drive.grand-challenge.org/DRIVE/>
4. **Messidor-2 — ADCIS**
<https://www.adcis.net/en/third-party/messidor2/>

5. **e-ophtha — ADCIS**

<https://www.adcis.net/en/third-party/e-ophtha/>

6. **DDR Dataset**

<https://github.com/nkicli/DDR-dataset>

Research Conclusion

The dataset analysis shows that **SIH26038 should be approached as a multi-task, multi-dataset medical computer-vision problem rather than a single image-classification problem.**

The strongest core combination is:

APTOS 2019 + IDRiD + DRIVE + Messidor-2

APTOS provides the foundation for five-level DR grading, IDRiD provides Indian-domain and lesion-level information, DRIVE provides vessel segmentation annotations, and Messidor-2 provides an opportunity for independent external evaluation.

This dataset strategy directly supports the major technical requirements of SIH26038 while keeping the system modular and scientifically defensible.